Perspicio: Safeguarding Against Non-Consensual Photo Sharing Over Social Networks

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Abstract—**Photo sharing over social networks has become part of daily life. However, such practices can inadvertently lead to privacy breaches, particularly when done without the consent of unaware individuals in photos, known as the non-consensual photo sharing issue. Existing strategies, such as interactive schemes and access policy-based mechanisms, often fail to effectively determine if individuals in a photo are knowingly captured. To address this, we propose *Perspicio*, a new system to protect users' privacy when their images are captured unintentionally. *Perspicio* not only ensures heightened privacy but also accurately detect the unawareness of individuals captured in photos. This is achieved by first integrating more specific features for personal identification and introduces a dynamic weighting approach to adjust the significance of these features based on different scenarios and individual characteristics. Second, it leverages a new protocol that enhances both authorization and sharing with a higher privacy level. Finally, we conduct extensive evaluations and demonstrate that it achieves outstanding performance. Compared with the state-of-the-art in identifying unaware individuals in photos, *Perspicio* boosts the prediction accuracy by up to 13.2%, while speeding up the processing by 862× and 1,425× at online and offline phases, respectively.**

Index Terms—**Non-consensual photo sharing, privacy protection, feature extraction, attention mechanism, social networks.**

I. Introduction

Camera-equipped devices are broadly deployed in many premises and spur the ubiquitous photo capture and sharing in our daily life. With the convenience of the Internet, these photos are widely disseminated over social networks. Two popular social networking service platforms, Instagram [1] and Facebook [2], were reported to have more than 1,000 and 40,000 images uploaded per second, respectively. The unprecedented surge in photo capture and sharing behaviors significantly increases the risk of privacy breaches. To be specific, the habit of indiscriminate life moment sharing causes individuals to be unconsciously photographed all the time. Even cautious individuals may become the subjects of photos taken by others, either intentionally (i.e., secretive photographing) or unintentionally (i.e., as passive observers). These are uniformly termed as non-consensual photo sharing issues in this paper. These photos depict unaware subjects and, if shared publicly, may inadvertently reveal their social information, locations, personal habits, and interests [3], [4], [5], [6]. Numerous privacy breaches are occurring worldwide due to photo sharing on social networks. For instance, Italian government departments receive more than 500 complaints regarding candid filming involving over 1,400 people in 2022 alone, nearly 80% increase from the previous year [7]. According to U.K. police data, there are 1,185 cases of private photos and videos being shared online in the same year. Meanwhile, 9,233 instances of personal residence trespassing occurs due to photo sharing in Germany [8]. Adding to the concern, U.K. media reports a massive 329% increase in cases of private photography in London alone, made worse during the pandemic [9]. Therefore, handling the non-consensual photo sharing issues is crucial to protect the privacy of individuals who do not realize that their photos are being shared.

Previous efforts to mitigate privacy breaches for individuals appearing in shared photos can be broadly categorized into three main types. The first type, *user-driven privacy protection*, requires active user participation, such as installing software to block unauthorized photo capture and sharing [10], [11], [12],

Received 16 October 2024; revised 14 May 2026; accepted 8 June 2026. Date of publication 19 June 2026; date of current version 1 September 2026. The work of X. Qian was also supported in part by the Sichuan Provincial Natural Science Foundation under Grant 2026NSFSC1466 and in part by the China Postdoctoral Science Foundation under Grant BX20250389. The work of G. Xu was supported in part by the National Natural Science Foundation of China under Grant 62502075 and in part by the Sichuan Provincial Natural Science Foundation under Grant 2026NSFSC0431. The work of H. Li was supported in part by the National Natural Science Foundation of China under Grant U25B2079 and in part by the Chengdu Science and Technology Program under Grant 2023-XT00-00002-GX. The work of Y. Fang and H. Cao was supported in part by the Hong Kong SAR Government under the Global STEM Professorship and Research Talent Hub and in part by the Hong Kong Jockey Club under the Hong Kong JC STEM Lab of Smart City under Ref. 2023-0108). This work was supported by Hong Kong Jockey Club Charities Trust was conducted in the JC STEM Lab of Smart City under Contract 2023-0108. *(Corresponding author: Guowen Xu.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TDSC.2026.3705636, provided by the authors.

Digital Object Identifier 10.1109/TDSC.2026.3705636

[13], [14], [15], [16], [17]. For example, Xu et al. [15], [18], [19] designed a facial recognition engine based on distributed machine learning (now known as federated learning) to protect individuals' privacy when sharing photos on social networks. However, these methods are limited by their reliance on user intervention and are vulnerable to evasion techniques. The second type, *server-based policy enforcement*, uses cloud servers to enforce access control policies [20], [21], [22], [23], [24], [25], [26]. These policies aim to restrict unauthorized access to photos. However, most policies are difficult to define before a photo is taken [20], [25], and many do not support fine-grained access control [22], [23]. The final type, *machine learning-based solution*, uses algorithms to identify unaware individuals in photos [27]. While promising, this solution is limited by its detection range, computational demands, and the risk of privacy exposure during consent querying processes. More details can be found in Section VII.

Our contributions: To tackle the challenges, we present *Perspicio*,[1] a novel non-consensual photo sharing system that accurately detects individuals in photos who are unaware of being photographed and efficiently shares photos with reduced risk of privacy leakage under the stated system assumptions. *Perspicio* has several innovations. First, *Perspicio* introduces a highly flexible feature fusion strategy, which carefully picks and integrates a variety of computer vision features that play a crucial role in identifying individuals who are unaware of being photographed. This enhances the system's accuracy across diverse scenarios. Second, *Perspicio* incorporates an advanced attention mechanism for feature fusion strategies, which assesses the importance of various features based on the actual context of a photo, allowing for dynamic weighting accordingly. This dynamic adaptation is particularly effective in complex or ambiguous situations, significantly improving the recognition accuracy. Third, *Perspicio* implements a rapid partial image encryption protocol, which encrypts only the privacy-sensitive elements (such as faces) within a photo, substituting these sections with ciphertexts, so that only authorized users can decrypt and view this information for approval.

Our major contributions are summarized as follows.

- We present *Perspicio*, a system with a multi-featured attentional integrator meticulously designed to identify unaware individuals in photos. Informed by our user study insights, this integrator harnesses five pivotal features through a dynamic weighting strategy tailored to *Perspicio*'s attentional mechanism.
- We design an advanced authorization mechanism for *Perspicio* that prioritizes user privacy in photo authorization and sharing. Central to this is an innovative rapid partial image encryption protocol to accelerate encryption by only encrypting sensitive image regions. This targeted encryption enables fine-grained access control, allowing nuanced management of users' photo privacy tailored for their specific needs and contexts.
- We conduct a real-world user study that identifies five key features for characterization and their context-dependent salience, and, to enable comprehensive evaluation, we compile a new five-scenario dataset (daily life, art, meeting, hospital, public transit) with per-person awareness labels from online sources. Extensive experiments show that *Perspicio* attains 88.2% accuracy (F1 = 0.881), surpassing state-of-the-art methods by 13.2% and reducing computational cost by $862\times$ (online) and $1425\times$ (offline), with overheads of only 1.14,ms and 0.68,ms, respectively.

[1]The Latin word for perceive and observe.

II. Problem Statement

The aim of *Perspicio* is to offer efficient and automated identification of non-consensual photo sharing. In this paper, awareness denotes a photographed individual's apparent awareness of being photographed at the moment of image capture, as inferred from observable visual cues and scene semantics. Accordingly, unaware individuals are those who do not exhibit such apparent awareness under this operational formulation. *Perspicio* enhances privacy protection by accurately and efficiently detecting such individuals in social-network photos.

A. Limitations of Previous Research

L1: **Automated Detection.** Almost all of the previous works fail to provide intelligent privacy protection in real-world social networks because they assume that all users are part of the system and implement privacy-preserving protocols only for internal users [10], [11], [12], [13], [14], [15], [16], [17], [20], [21], [22], [23], [24], [25], [26]. However, real social networks are highly complex and often include unregistered users. For these external groups, new methods must be introduced to safeguard user privacy. One possible approach is to protect user privacy by detecting whether individuals in photos are aware of being photographed [27]. While this method can partially mitigate privacy risks for external groups in an automated manner, its "automated" capability is limited and heavily influenced by the environment and context. Specifically, it faces challenges in two aspects: feature selection and classifier adaptability. The former relies on detecting patterns based on only two simple features, and this narrow feature set is insufficient to address the diverse nature of photo scenes. The latter struggles to adapt to various scenarios, as individuals in photos often exhibit different reactions depending on the context.

L2: **Privacy and Efficiency.** The privacy concerns associated with photo sharing on social networks stem from two key factors. First, current advanced solutions are vulnerable to identity fraud attacks, where malicious adversaries exploit registration loopholes to create fake identities, resulting in potential privacy breaches in shared photos [20], [22], [27]. Second, it is difficult to prevent malicious users on large-scale social platforms from accessing others' photos, as each photographed user is granted access to the entire group photo during the authorization phase [27], [28]. On the other hand, these privacy protection schemes involve high computational costs during the user authorization process due to the complexity of cumbersome cryptographic techniques [27], which poses significant challenges for social media platforms managing large volumes of photo uploads every second.

B. Key Insights and Ideas

In this section, we summarize three main insights and ideas that form the core and essence of building *Perspicio*. The first two aim to address the limitations of *automated detection* (***L1***), while the third addresses *privacy and efficiency* issues (***L2***).

1. We design a multi-feature fusion classifier (MFFC) that integrates various visual and contextual cues indicative of apparent unawareness, enabling the detection of individuals who do not appear to be aware of being photographed across a broader range of scenarios: In our study, the primary challenge is to accurately distinguish individuals who appear unaware of being photographed from those who appear aware across diverse scenes. This task is difficult because apparent unawareness is inherently subtle, subjective, and context-dependent, especially in environments such as daily-life photos and artistic shots. To address this, we conduct a user study to examine human perceptions across different scenes, where target individuals in photos are judged as "unaware" or "aware" according to participants' social norms, experiences, visual observations, and scene understanding. Our focus was on understanding how scene context influences judgments of unawareness and the identification of relevant visual cues. When comparing human observers to automated detectors, we found that humans integrate non-visual information and intuition, while detectors rely solely on visual data. Participants highlighted distinct visual features associated with apparent unawareness in various scenarios, demonstrating significant variability. This study reveals that recognizing apparent unaware states varies widely across different contexts, with each feature's importance fluctuating from scene to scene. Therefore, in multi-scene detection, it is crucial to consider a diverse set of features rather than relying on a limited two-feature approach.

2. We propose a novel attentional mechanism for our MFFC, allowing it to effectively modulate the importance of different features across various scenarios: In our study, we aim to address the complex challenges presented in identifying unaware states in diverse settings. To be specific, variations in behaviors across indoor and outdoor, static and dynamic, and crowded and sparse environments add complexity to the task. Moreover, interpreting visual features like gaze direction, head orientation, emotions, and body posture becomes more complex in different contexts, necessitating dynamic adjustments for improved detection accuracy and applicability. To address these challenges, we create a new dataset to encompass a broader range of scenarios, including daily life, art, meeting, hospital, and public transit, with each one carefully annotated. With a focus on visual features indicative of unaware states, we develop multiple classifiers based on both individual and fused features. In addition, we implement a dynamic weighting mechanism and seamlessly integrate it into our effective classifier to adapt to scene variations (Sections IV-A).

3. We construct an efficient authorization mechanism that enables fine-grained access control over photo privacy during the photo sharing process, while ensuring efficiency: Our study addresses the challenge of enhancing the efficiency of photo sharing while maintaining privacy. The core issue is that previous solutions only protect privacy after photos have been shared, overlooking the significant privacy risks that arise even during the pre-sharing authorization phase. Directly sharing photos without adequate protection can result in severe security vulnerabilities. While public key encryption seems like an intuitive solution, encrypting entire images significantly increases computational and communication overheads and does not allow for precise protection of each individual's photo information. To overcome these limitations, we depart from the cumbersome biometric-based authentication approach and propose a more refined solution. We partially encrypt photos using a combination of traditional encryption techniques, integrating public key encryption with symmetric cryptography. This method provides strict access control over private photo information while minimizing overheads. Additionally, by incorporating a digital signing authorization method, we develop an efficient and secure photo sharing authorization scheme, as detailed in Section V-A.

III. Background and Motivation

A. Cryptographic Background

1) Hybrid Encryption: Hybrid Encryption combines the security benefits of asymmetric key exchange with the efficiency of symmetric encryption for the transfer of large volumes of data. The hybrid encryption system employed in *Perspicio* utilizes Elliptic Curve Diffie-Hellman (ECDH) for the key exchange process. Through this process, the participating parties generate a "shared secret". This shared secret is not directly used as the encryption key, and instead, it serves as the foundation for deriving a session key through a Hash-Based Key Derivation Function (HKDF) [29], which is then used for the AES encryption and decryption of messages. This methodology ensures that the communication is securely encrypted, distinguishing clearly between the process of securely sharing a secret (key exchange) and the process of encrypting and decrypting data using the derived session key. The detail is given as follows.

- ***Key Generation:*** The key generation algorithm, upon receiving the security parameter 1^λ as input, produces a secure and corresponding public-private key pair (pk, sk) for the user:

$$(pk, sk) \leftarrow \mathsf{ECDH.KeyGen}(1^\lambda).$$

- ***Key Exchange:*** Elliptic Curve Diffie-Hellman algorithm (ECDH) is used to securely exchange keys between two parties over an insecured channel to obtain a shared secret (SS):

$$\mathsf{SS} \leftarrow \mathsf{ECDH}(pk_A, sk_B) = \mathsf{ECDH}(pk_B, sk_A).$$

- ***Symmetric Encryption:*** Once the key is exchanged, this shared secret is derived into the key K (by HKDF) for a symmetric encryption algorithm such as AES (Advanced Encryption Standard) to encrypt and decrypt messages:

$$\mathsf{ct} \leftarrow \mathsf{AES.Enc}(K, \mathsf{Msg}), \mathsf{Msg} \leftarrow \mathsf{AES.Dec}(K, \mathsf{ct}).$$

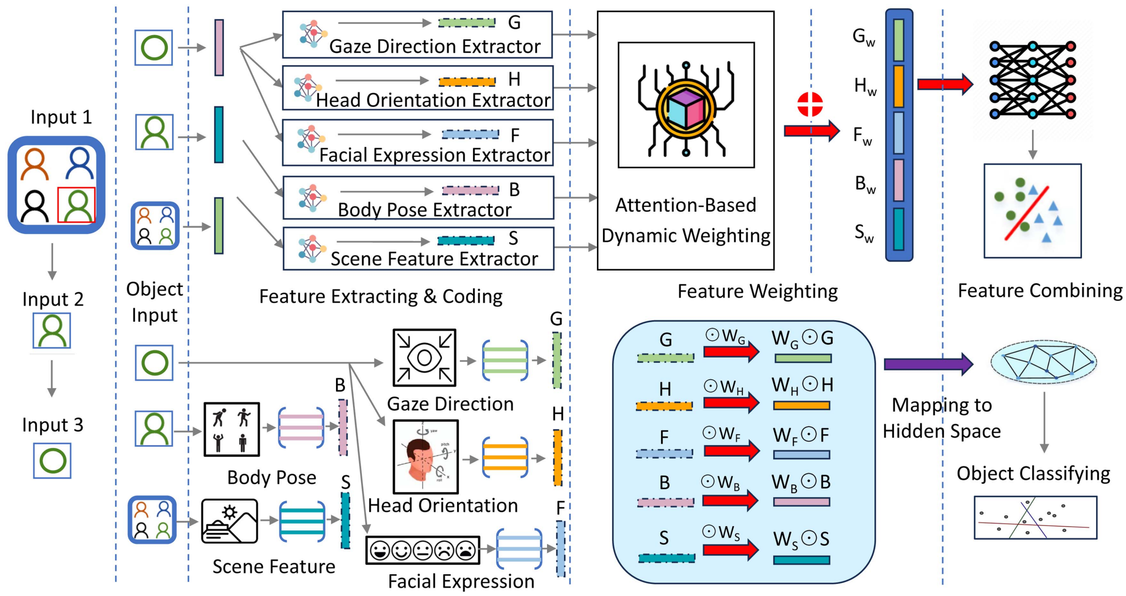


Fig. 1. Overview of Multi-Feature Attention Integrator (MFAI) and MFAI Interpreted from feature extracting, weighting, and combining.

The key K mentioned above can be updated using an old key K' as follows: $K \leftarrow \mathsf{HKDF}(K')$.

An Illustrative Example: To make the following protocol more understandable, we provide an intuitive example. Suppose Alice wants to share a photo with Bob. They first use ECDH to agree on a shared secret over an insecure channel. This secret is then transformed into a session key through HKDF. With this key, Alice encrypts only the sensitive region (e.g., Bob's face) using AES. Finally, Bob, holding the corresponding private key, can decrypt and view his facial area.

2) Signature: Let λ be the security parameter of the scheme, $\mathcal{M}$ and $\mathcal{S}$ be the sets of messages and signatures, respectively. A signature scheme is given by three probabilistic polynomial time algorithms:

- $\mathsf{Sign.KeyGen}(1^\lambda) \rightarrow (vk, sk)$: takes as input the security parameter λ and outputs a pair of keys, the verification key vk and the signing key sk.
- $\mathsf{Sign}(1^\lambda, sk, M) \rightarrow \sigma$: takes as input the security parameter λ, the signing key sk, and a message $M \in \mathcal{M}$, and outputs a signature $\sigma \in \mathcal{S}$.
- $\mathsf{Verify}(1^\lambda, vk, M, \sigma) \rightarrow \{\text{accept, reject}\}$: takes as input security parameter λ, verification key vk, message M, and signature σ, and either accepts or rejects.

The signature is correct if for all message $M \in M$, $(vk, sk) \leftarrow \mathsf{Sign.KeyGen}(1^\lambda)$, then $\mathsf{Verify}(1^\lambda, vk, M, \mathsf{Sign}(1^\lambda, sk, M)) = \text{accept}$ with overwhelming probability. In this paper, we require the signature scheme be selective unforgeability against chosen message attack (SU-CMA) secure.

B. Motivation

We conducted a user study (see Appendix IX-B,) to analyze the current deficiencies of non-consensual photo sharing systems and the underlying reasons. This research provides sufficient preliminary evidence for the informed design of our new system. The findings of the study can be summarized as follows.

- **F1:***Awareness-determining features in photos vary greatly across scenarios.*
- **F2:***Scenario context significantly affects the relevance of awareness-signifying features in photos.*
- **F3:***Five key features, namely, gaze direction, head orientation, body language, scene context, and facial expression, are crucial to detecting a person's awareness in photos.*

Accordingly, under the definition in Section II, the role of the user study is not to measure a psychologically exact or directly verifiable internal mental state. Instead, it provides empirical grounding for training a context-aware machine learning model that approximates how human observers infer a photographed individual's apparent awareness of being photographed from visible cues in a scene. In this sense, the corresponding labels used in *Perspicio* are better understood as per-person, context-grounded judgment labels for system design, training, and evaluation, rather than as direct measurements of an individual's true internal mental state. Based on ***F1*** and ***F2***, we design a novel Multi-Feature Attention Integrator, which leverages the five most pivotal features for awareness detection (as identified in ***F3***), dynamically weighs their importance across scenarios, and then performs classification accordingly.

C. Threat Model and Design Goal

Threat Model: Our threat model comprises four entities: the photo sharer, the user, the cloud server, and a trusted third party. The last entity is widely accepted in the real world [30], [31]. The descriptions of these four entities within the threat model are as follows.

- ***Photo-Sharer.*** We assume that the photo sharer is semi-honest, who uploads images in order to share and obtains authorization from the individuals in photos.
- ***User.*** Any user in the system are considered malicious, who can launch any attacks to make privacy leakage of other users.
- ***Cloud Server.*** The cloud server is assumed to be semi-honest, which exactly follows the protocol and will not collude with any other entities.
- ***Trusted Third Party.*** The Trusted Third Party (TTP) is fully trusted in the current threat model and serves only as a trust anchor for key generation and initialization only at registration time, i.e., generating and distributing the keys for all parties and binding registration information to the corresponding key materials. It is not involved in the subsequent online stages, such as photo preprocessing, feature extraction, awareness detection, authorization decisions, or final sharing decisions.

Unless otherwise stated, stronger adversarial settings, such as malicious cloud deviations, TTP compromise, or cross-entity collusion, are out of the scope of this work.

Design Goal: We focus on the following two primary design goals: (1) developing an automated detection method for unaware individuals applicable in a wide range of scenarios, aimed at enhancing the accuracy and reliability of detecting unaware states in diverse environments; (2) constructing a secure photo sharing authorization scheme using basic cryptographic primitives. This scheme is designed to maximally protect privacy during the photo upload, authorization, and sharing processes, thereby preventing data leakage to some extent. Our design goals strive to bridge the gap between technological innovation and practical applications, combining accurate unaware state detection techniques with efficient privacy measures to ensure comprehensive protection of user privacy during photo sharing over social networks.

IV. User-Driven Multi-Feature Attentional Integrator

Based on the findings from our user study, we construct the Multi-Feature Attentional Integrator (MFAI), the combination of a Multi-Feature Fusion Classifier (MFFC), and Dynamic Weighting with Attentional Mechanism (DWAM). It integrates a rich set of human-related features along with specialized scene features, thoroughly considering the character and scene information in photos. Through the attention mechanism, it dynamically allocates weights to different types of features, aiming to enhance the generalization of classification tasks.

A. Multi-Feature Fusion Classifier

Fig. 1 shows the construction process of MFAI. Firstly, the uploaded images undergo object detection to extract necessary characters and scenes. For character feature extraction, we need to crop each character image to obtain the head and body regions, respectively, which will serve as different inputs for backbone networks. Additionally, we employ the same method to extract specialized scene features from complete photos. The model used in our design and how we extracted features are presented next.

Head-related feature extraction: Our head-related feature contains three parts as discussed below.

Gaze direction. We utilize the pre-trained Gaze360 model [32] on a large dataset to extract a person's gaze direction. This model processes a sequence of 7 frames to predict the gaze in the central frame. It first applies ResNet18 [33] to process the target person's head to generate a 256-dimensional feature, which is then fed into LSTM [34] and a fully-connected layer to produce gaze prediction and error quantile estimation. We input cropped facial images from our dataset into this model and extract features from the penultimate network layer for our classifier.

Head orientation. We utilize the image2pose model [35] pre-trained on the AFLW2000-3D dataset [36] to deduce head orientation. The image2pose framework operates on a two-tier network rooted in Faster R-CNN [37]: the initial tier encompasses a region proposal network (RPN) allied with a feature pyramid network (FPN), tasked with pinpointing prospective face locales; the subsequent tier employs region of interest (ROI) pooling to harvest features, channeling them to two distinct head prediction tasks. Cropped facial images of individuals from our dataset are input into the ROI pooling and the fully-connected layer, procuring the final face pose which serves as the requisite head orientation feature.

Facial expression. We utilize a facial emotion recognition model pre-trained on the FER2013 dataset [38] to extract a person's facial expression. This is a fine-tuned model based on ResNet18 with 73.7% accuracy. Similar to the extraction of gaze direction and head orientation, we input the cropped facial images into this model and extract facial expression features from the feature layer.

Body feature extraction: We use OpenPose [39] to extract the body pose feature of a person, which aims at detecting 18 regions of human body and outputs joints along with detection confidence. Given the abundant feature layer parameters (27,000) in the OpenPose model, we utilize the Principal Component Analysis (PCA) algorithm to compress them into a vector of length 512 for efficiency enhancement and feature standardization.

Scene feature extraction: We pre-train the model on our constructed dataset by manually annotating scene labels and using VGG16 [40] for training. A feature extraction layer is added before the final classification layer to capture scene-related features. During inference, the entire photo is fed to the pre-trained model, and the penultimate layer's output provides the scene feature vector.

B. Dynamic Weighting With Attention Mechanism

We utilize the attention mechanism for dynamic weighting of the features extracted from the aforementioned backbone networks. This mechanism mimics the internal process of biological observation behaviors, i.e., aligning internal experiences with external sensations to enhance the observation precision of certain areas.

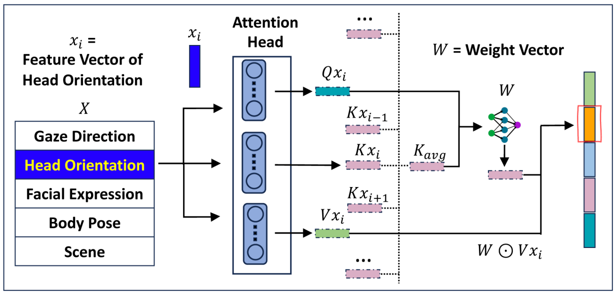


Fig. 2. Attentional dynamic weighting mechanism.

Our dynamic weighting method is shown in Fig. 2. Each feature input is transformed into 64-dimensional query (q), key (k), and value (v) vectors. We then concatenate the q and v vectors of the same feature, and take the mean of k vectors from all features as the input to the attention model. The model outputs a vector of equal dimension, which is combined with the original 64-dimensional value vector through the Hadamard product (denoted as $\odot$ in Figs. 1 and 2), i.e., $W \odot V$, to obtain dynamically weighted feature values. This dynamic weighting can be performed in parallel, and afterward we concatenate all the weighted vectors.

Through the implementation of the attention mechanism and feature-level fusion, the model can achieve dynamic weighting and compact representation of features, thereby enhancing the accuracy in non-consensual classification or detection tasks as well as the model interpretability. These techniques further bolster the model's capabilities of adaptive learning, generalization, and robustness. By integrating features from multiple sources, we improve the model's performance across various scenarios and datasets. Finally, we train an MLP classifier using fused features and the established dataset. It is worth noting that the proposed dynamic attention mechanism inherently enhances robustness under uncontrolled conditions. Specifically, when facial features (e.g., gaze or expression) are unreliable due to occlusion, poor lighting, or partial visibility, the mechanism adaptively reallocates attention weights towards other available cues such as body posture and scene semantics. This redundancy allows *Perspicio* to maintain reliable performance even in challenging in-the-wild scenarios where certain feature modalities may be degraded or missed.

V. Perspicio

From a high-level perspective, our *Perspicio* comprises two main components: a multi-feature attentional integrator that automatically identifies unaware individuals in photos and a secure authorization mechanism that validates the authenticity of the photo sharing consent information in a cryptographic approach. Next, we provide the detailed design of our *Perspicio* with the integration of these two components.

A. System Design

Fig. 3 shows the workflow of our *Perspicio*. We describe the detailed design as follows. ***Registration.*** Each user registers

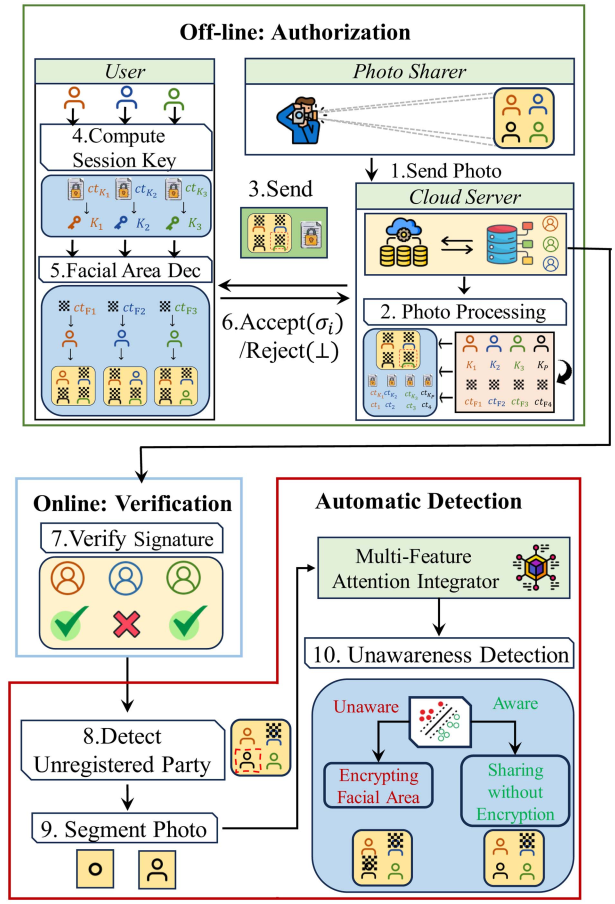


Fig. 3. The Workflow of *Perspicio*.

Perspicio with his/her name and face information, which must be captured in real-time using a camera to gather two-dimensional facial features. To ensure the authenticity of the facial data during the registration, *Perspicio* employs an efficient liveness detection method based on the widely-used "challenge-response" technique, guaranteeing the registration of a live face rather than through a photo. Subsequently, *Perspicio* requests TTP to execute the ECDH.KeyGen and Sign.KeyGen algorithms, generating two pairs of public and private keys for encrypting and authenticating the user's facial information. The private keys sk_i^{ecdh} and sk_i^{sig} are securely delivered to the corresponding user U_i with the corresponding public keys pk_i^{ecdh} and pk_i^{sig} publicly available. *Perspicio* also obtains a pair of keys (pk_P, sk_P). Regarding data storage, we extract feature representation vectors from the detected live face images and store them using hybrid encryption (ECDH + AES) combined with digital signatures to ensure confidentiality and integrity against unauthorized access or tampering.

Photo Preprocessing. The cloud proceeds to segment each face, perform facial recognition, and correlate it with a specific ID_i. If the detected face belongs to the photo publisher, *Perspicio* assumes implicit consent to share and therefore does not apply encryption to that face. For all other registered users, the cropped facial image is encrypted by invoking the AES.Enc function with

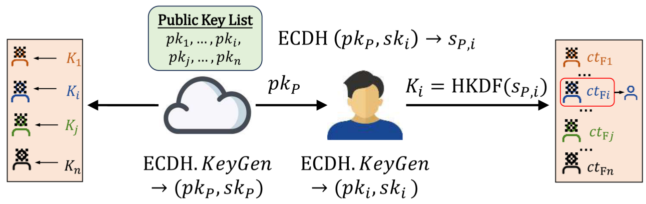


Fig. 4. Illustration of selective encryption using hybrid encryption.

session key K_i or K_i', where $K_i = \mathsf{HKDF}(\mathsf{ECDH}(pk_i, sk_P))$ or $K_i' = \mathsf{HKDF}(K_i)$. The faces of all unregistered users in the photo are also encrypted using the session key $K_P = \mathsf{HKDF}(\mathsf{ECDH}(pk_P, sk_P))$. This partially encrypted photo is then distributed to the corresponding party with a specific box.

Authorization. In this phase, *Perspicio* requests the authorization of user U_i. U_i first computes session key K_i or K_i' by using $\mathsf{ECDH}(pk_P, sk_i)$ or HKDF, and then decrypts the boxed area of the photo by implementing the $\mathsf{AES.Dec}$ algorithm to recover the encrypted facial area, as shown in Fig. 4. Upon recognizing his/her own presence in the photo, he/she decides whether to share it with others. The user who chooses to share the photo must invoke the Sign algorithm to generate a signature σ_i for the recovered facial photo (FP_i) to complete authorization. Subsequently, the pair $(\mathsf{ID}_i, \sigma_i)$ is appended to the photo's metadata. This is done after receiving signatures from all the intended recipients, which include a subset of registered users contained in the photo. In the case where individuals do not wish to share the photo, they can opt not to sign and return $(\mathsf{ID}_i, \perp)$.

Verification. *Perspicio* extracts the photo's metadata and verifies the signature by calling the Verify algorithm for U_i. A successful verification indicates U_i has authorized to share this photo. If the verification fails or there is no signature for U_i, *Perspicio* masks the facial area with the encrypted data to protect the privacy.

Automatic Detection. Based on the multi-feature classifier and importance of each feature in different scenarios, we build a multi-feature attentional integrator to detect unaware individuals automatically. Those individuals who do not register with *Perspicio* will be fed to our integrator for detection. The head area of unaware users detected will be encrypted while that of aware users will not be altered.

Photo Sharing. *Perspicio* processes the photos and then shares them, taking into account the user authorization states and the integrator's detection results, in order to minimize the risk of unauthorized photo capture to the greatest extent possible.

For a comprehensive illustration of the full workflow in a realistic photo-sharing scenario involving multiple registered and unregistered users, please refer to the detailed user case presented in Appendix IX-E, available online.

B. Security Analysis

In this section, we conduct the security analysis for *Perspicio* under the threat model in Section III-C. Our goal is to clarify the privacy and authenticity properties supported by the current design. In particular, the following analysis assumes a trusted third party (TTP) during initialization and a semi-honest, non-colluding cloud server during protocol execution. Stronger settings, such as malicious cloud server, TTP compromise, or cross-entity collusion, are out of the scope of this paper.

Definition 1 (Photo Privacy). The photo privacy of a registered *Perspicio* user is preserved if, for any probabilistic polynomial-time attacker in the adopted threat model, given the security parameter λ, a session key K_i, and an initialization vector IV, the AES-CTR ciphertext corresponding to the protected facial region of that user is computationally indistinguishable from a random bit string of the same length.

Definition 1 formalizes the core confidentiality goal of *Perspicio* for registered users. Based on this notion, we next state a broader system-level security claim. In particular, Property (2) below directly realizes Definition 1, whereas Properties (1) and (3) provide the guarantees required for securely enforcing this privacy goal in the full protocol. Property (4) further extends protection to unregistered users, but this extension is only conditional on detector correctness.

Theorem 5.1 (Perspicio System Security). Assume that ECD H is secure for shared-secret establishment, HKDF is a secure key-derivation function, AES-CTR is semantically secure under the derived keys, and the signature scheme is SU-CMA secure. Then, under the stated threat model, *Perspicio* provides: (1) *secure identity-to-key binding during registration;* (2) *confidentiality of encrypted facial regions for registered users, achieving Definition 1;* (3) *authenticity of authorization decisions for registered users;* and (4) *detector-guided privacy protection for unregistered users.*

Proof. We prove Theorem 5.1 by showing that Property (2) achieves Definition 1 for registered users, with Properties (1) and (3) ensuring that this privacy goal is correctly bound and enforceable within the protocol. Property (4) extends protection to unregistered users in a detector-guided but conditional manner.

During registration, *Perspicio* requires liveness-verified facial capture before key provisioning. The TTP generates sk_i^{ecdh} and sk_i^{sig}, securely delivers them to U_i, and binds the corresponding public keys to ID_i. Under the trusted-TTP assumption, this establishes secure binding among user identity, liveness-verified registration information, and key materials. During photo preprocessing, the facial region of each registered user is encrypted under

$$K_i = HKDF(ECDH(pk_i, sk_P)), \tag{1}$$

or the updated key $K_i' = HKDF(K_i)$. Since only the corresponding private key material sk_i^{ecdh} enables derivation of the same shared secret, unauthorized parties cannot recover the protected facial region. By the security of ECDH, HKDF, and AES-CTR, the ciphertext remains computationally indistinguishable from random to any unauthorized adversary in the adopted model.

During authorization, user U_i decrypts the protected facial region, obtains FP_i, and decides whether to authorize sharing. If the consent is granted, U_i generates a signature σ_i on FP_i, and the pair (ID_i, σ_i) is appended to the photo metadata. During verification, *Perspicio* validates σ_i using the verification key bound to ID_i. Since sk_i^{sig} is bound to ID_i during registration and the signature scheme is SU-CMA secure, no adversary without sk_i^{sig} can forge a valid authorization on behalf of U_i.

For unregistered individuals, *Perspicio* encrypts their facial regions using $K_P = HKDF(ECDH(pk_P, sk_P))$. Whether such regions remain encrypted in the final shared photo further depends on the output of the multi-feature attentional integrator. Therefore, unlike the registered-user case, the privacy protection for unregistered users is conditional on detector correctness rather than being purely cryptographic.

Overall, *Perspicio* provides registration-time binding, encrypted facial-region confidentiality, and authorization authenticity under the adopted threat model.

Remark. However, we do not claim that these guarantees can be extended to malicious protocol deviations, TTP compromise, cloud-user collusion, or other stronger active adversaries. Likewise, the protection of unregistered users is not independent of classification accuracy. These stronger settings are left for future work.

VI. Evaluation

In this section, we evaluate our *Perspicio* and compare its functionality with the most relevant works [5], [11], [11], [13], [14], [16], [17], [20], [21], [22], [23], [25], [26]. Given that our approach utilizes machine learning techniques, it is most closely related to *Videre* [27]. Therefore, we provide a detailed comparison with *Videre* in terms of prediction performance, computational cost, and communication overhead. We emphasize that the following evaluation provides prototype-level empirical validation rather than a production-stage, end-to-end deployment study on a commercial social-networking platform. Specifically, our results are based on an implementation of the core system pipeline using real models and cryptographic modules, evaluated on a real dataset and a local computing testbed. Factors such as real platform integration, networked cloud execution, large-scale concurrency, and operational service orchestration are beyond the scope of the current evaluation.

A. Experimental Setup

We implement our *Perspicio* using Python, leveraging various pre-trained models such as the gaze prediction model [32], head orientation prediction model [35], facial expression recognition model [41], and body language extraction model [42]. Additionally, we train a scene prediction model employing a VGG16 architecture [43]. The implementation of cryptographic parts is also in Python, utilizing Pycryptodome [44] library that facilitates Elliptic Curve Digital Signature Algorithm (ECDSA) [45] and hybrid encryption [46] based on Elliptic-curve Diffie–Hellman (ECDH) [47] and AES-CTR [48]. All experimental outcomes are obtained on a workstation equipped with the 12th Gen Intel(R) Core(TM) i9-12900H CPU at 2.50 GHz, 32.0 GB RAM, and an NVIDIA 4060 GPU running Linux.

B. Dataset Description

Due to the incapability of the awareness dataset from the [27] scheme to precisely classify unawareness in specific scenes, a prerequisite for this task is to collect data for different scenes. Therefore, we compile an abundant awareness dataset and manually annotated the levels of awareness to facilitate a more nuanced understanding and accurate classification across various scenarios. This significantly improves the model's ability to generalize across different contexts, thereby contributing to the robustness and effectiveness of our *Perspicio* in real-world applications. Following the operational definition in Section II, the corresponding awareness labels are treated as per-person, context-grounded judgment labels rather than psychologically exact ground truth. They are constructed from observable visual cues and scene semantics, and are used to train and evaluate a context-aware machine learning model that approximates how human observers infer apparent awareness in a scene. This design improves the practical realism of the dataset, while making it explicit that the labels may still be affected by subjective interpretation, conflicting visual evidence, and ambiguity in borderline cases. Moreover, uncertainty is not omitted in the current study. In our user study, participants were asked to judge awareness using a graded response scale, including "Not sure", so perceptual uncertainty was explicitly elicited at the judgment stage rather than being ignored from the outset.

Dataset Construction: To construct our dataset, we initially engage three individuals to independently gather a total of 4,423 sample photos from various websites, search engines, and real social platforms, encompassing a wide array of scenes, text contents, and crowds. Importantly, many of these images naturally contain challenging conditions such as occluded faces, poor illumination, or partially visible individuals. We deliberately retained such imperfect samples to better approximate in-the-wild photo-sharing environments. These were then combined with the original awareness dataset in [27]. To ascertain the status of each entity within the photos, we annotated the individuals depicted, with the dataset totaling 12,597 individuals.

Some photos feature multiple individuals. Taking into account common photo sharing scenarios and the clarity of the cropped images, we follow the dataset construction guideline in [27]: each photo should have no more than five individuals. This rule applies even in settings like meetings with numerous participants, where we only selected images showcasing up to five individuals. For the photos with N individuals, we adopt a methodology similar to [27], duplicating the individual images N times and sequentially marking the individuals using bounding boxes.

In an effort to maintain consistency with prior dataset construction processes in [27], we employ crowdsourcing techniques to manually annotate whether the entities in photos are aware they are photographed. We further harness immersive enhancement technology to swap the workers with the individuals in photos, and employ hypothetical questioning to foster empathy, thereby aligning the workers' subjective judgments with their genuine feelings. Ultimately, among 12,597 individuals, 5,935 (47.12%) are labeled as "aware party" while 6,662 (52.88%) are designated as "unaware party". For our experiment, the 12,597 individuals are split into a training set and a test set, consisting of 10,180 photos (80.82%) and 2417 photos (19.18%), respectively.

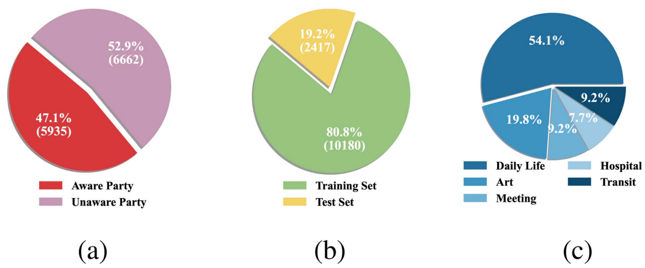


Fig. 5. Our Dataset Overview: Distribution of Individual Consciousness States and Dataset Split.

Scene Dataset: We use all images from our new dataset to create a fresh scene-based dataset (see Fig. 5), since no suitable existing dataset is available. Subsequently, we train a model to predict different scene categories, classifying images into five main types: daily life, art, meeting, hospital, and public transit, comprising 6,857 daily life photos, 2,502 artistic photos, 1,168 meeting photos, 1,099 hospital photos, and 971 public transit photos.

It should be specifically noted that due to the extensive scope encompassed by photos of daily life, there is a potential overlap with other scenes. Therefore, in our categorization of daily life scene photos, we deliberately exclude images from other four categories to ensure clear classification boundaries.

C. Ablation Study

1) Feature Combinations Under Different Models: In our study, we systematically assess the impact of different feature combinations on the performance of various machine learning models. The features considered are from our conducted user study in Section IX-B, including head orientation ($\mathcal{H}$), gaze direction ($\mathcal{G}$), facial expression ($\mathcal{F}$), scene features ($\mathcal{S}$), and body language ($\mathcal{B}$). We construct feature sets by combining these elements in various ways, resulting in subsets such as $\mathcal{HGFSB}$ (all features) and $\mathcal{HGFS}$ (all but $\mathcal{B}$), and test their impact on model accuracy and F1-Score. Several established machine learning models are employed in this analysis, including Decision Trees (DT), Random Forest (RF), AdaBoost, Support Vector Machines with linear and RBF kernels (e.g., SVM(Linear) and SVM(RBF)), Multilayer Perceptron (MLP), and our proposed model. Each model is evaluated on its ability to accurately predict outcomes in different scenarios: Daily Life, Art, Meeting, Hospital, Transition, and mixed scenes (ALL).

In Table I, we select the feature combinations under different combination sizes, specifically $\mathcal{GF}$, $\mathcal{GFS}$, $\mathcal{HGFS}$, and $\mathcal{HGFSB}$. To further assess the impact of scene features on the experimental outcomes, we also showcase the results of experiments incorporating scene features, such as $\mathcal{GS}$, $\mathcal{GFS}$, and $\mathcal{HGFS}$. The experimental results highlight the varying effectiveness of feature combinations across different scenarios and models. The $\mathcal{HGFSB}$ combination consistently deliver strong performance across nearly all models and scenarios. Our proposed model leveraging the full feature set shows a remarkable performance, achieving the highest accuracy and F1-Score in the "Art", "Meeting" "Hospital" and '‘Transition" categories. It also excels at the mixed scene scores, indicating the robustness of our model when all features are considered. It is noteworthy that the inclusion of more hierarchical features, especially "$\mathcal{S}$", in the combinations generally leads to improved performance, as seen with the $\mathcal{HGFS}$ and $\mathcal{HGFSB}$ subsets. This suggests that scene features play a critical role in enhancing model predictive capabilities. Our model outperforms traditional models in scenarios requiring nuanced understanding, such as "Art" and "Meeting", indicating its superior capability in handling complex feature interactions. The SVM with RBF kernel also performs notably well, especially with the $\mathcal{HGFSB}$ feature set, underscoring the importance of non-linear decision boundaries in complex feature spaces.

2) Feature Importance: We delve into the feature importance analysis across scenarios by analyzing their contribution to the outcomes of our model. In the prediction phase, the model's output of weight parameters is meticulously computed for five distinct features across a variety of scenarios. The statistical analysis is summarized in the box plots shown in Fig. 6, which help clarify how each feature is distributed and how important it is at various situations.

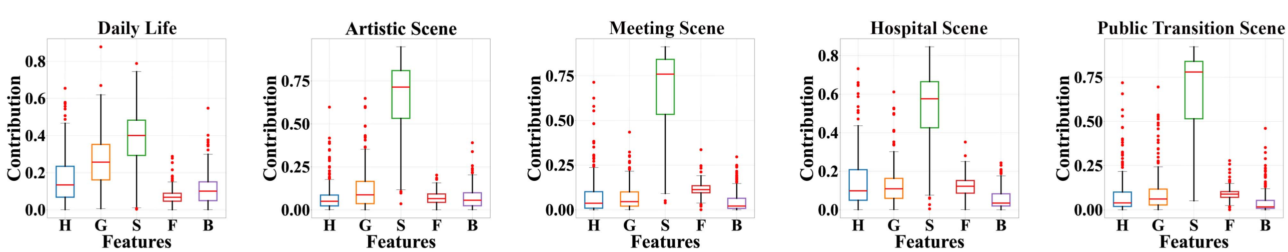
The scene feature $\mathcal{S}$ shows a significant dominance in its median value and range between the first and third quartiles across all scenarios, indicating its crucial predictive capability and fundamental role in attentional mechanisms. This widespread presence highlights the essential nature of context within visual perception, especially in dynamic environments like public transportation and clinical settings, where it determines the main focus of attention. Gaze direction $\mathcal{G}$, conversely, holds increased importance within daily life scene, as shown by its higher median and many outliers. This implies a central role in daily social interactions and activities, and where one looks can strongly convey their attention and purpose. Nonetheless, this aspect's significance decreases in situations where contextual scene details might matter more.

The impact of head orientation $\mathcal{H}$ and facial expressions $\mathcal{F}$ varies, adjusting their significance based on the situational context. For instance, in artistic scenes, gaze direction is more important, indicating the necessity to focus on visual cues. On the other hand, in meetings, facial expressions become more noticeable, emphasizing their role in communication and social interactions. Body language $\mathcal{B}$, although consistently having relatively weak statistical influence in all situations, should not be overlooked. Its subtle effects reflect the intricate aspects of non-verbal communications, which, despite their lesser importance overall, can be highly informative in specific nuanced interactions.

These findings emphasize the adaptability of visual attentional models and underscore the importance of thoroughly evaluating the significance of features in diverse situations. This understanding is crucial for improving feature selection and model training tailored for specific scenarios. Moreover, the observed differences in feature importance across various situations are consistent with our earlier discovery ***F2***.

TABLE I
COMPARISON OF ACCURACY AND F1-SCORE METRICS FOR SELECTED COMBINATION OF FEATURES AND MODELS

Features	Models	Daily Life		Art		Meeting		Hosptial		Transition		ALL	
		Acc.	F1-S.	Acc.	F1-S.	Acc.	F1-S.	Acc.	F1-S.	Acc.	F1-S.	Acc.	F1-S.
$\mathcal{GS}$	**DT**	0.718	0.714	0.751	0.300	0.537	0.469	0.719	0.711	0.636	0.642	0.645	0.645
	RF	0.814	0.812	0.292	0.132	0.681	0.659	0.836	0.872	0.764	0.815	0.697	0.706
	Adaboost	0.782	0.778	0.372	0.202	0.615	0.561	0.742	0.749	0.656	0.668	0.674	0.676
	SVM(Linear)	0.823	0.821	0.294	0.134	0.693	0.672	0.836	0.872	0.749	0.806	0.703	0.711
	SVM(RBF)	0.803	0.800	0.306	0.143	0.658	0.626	0.820	0.860	0.728	0.779	0.687	0.694
	MLP	0.826	0.825	0.320	0.155	0.693	0.674	0.820	0.847	0.754	0.818	0.709	0.718
	Ours	**0.833**	**0.835**	**0.284**	**0.442**	**0.681**	**0.705**	**0.820**	**0.781**	**0.744**	**0.688**	**0.704**	**0.695**
$\mathcal{GF}$	**DT**	0.718	0.714	0.751	0.300	0.537	0.469	0.719	0.711	0.636	0.642	0.645	0.645
	RF	0.824	0.822	0.823	0.742	0.817	0.825	0.852	0.884	0.769	0.845	0.820	0.821
	Adaboost	0.786	0.783	0.784	0.688	0.774	0.759	0.766	0.766	0.769	0.827	0.782	0.782
	SVM(Linear)	0.812	0.809	0.814	0.730	0.856	0.867	0.813	0.843	0.764	0.820	0.813	0.814
	SVM(RBF)	0.730	0.727	0.929	0.894	0.875	0.884	0.820	0.868	0.779	0.846	0.792	0.793
	MLP	0.775	0.773	0.829	0.751	0.887	0.889	0.844	0.856	0.826	0.839	0.805	0.805
	Ours	**0.805**	**0.808**	**0.784**	**0.879**	**0.848**	**0.825**	**0.820**	**0.788**	**0.800**	**0.750**	**0.806**	**0.805**
$\mathcal{GFS}$	**DT**	0.585	0.579	0.773	0.674	0.724	0.693	0.836	0.860	0.790	0.847	0.666	0.665
	RF	0.820	0.817	0.780	0.683	0.778	0.779	0.828	0.862	0.785	0.841	0.805	0.806
	Adaboost	0.690	0.686	0.762	0.759	0.864	0.855	0.836	0.847	0.815	0.837	0.740	0.741
	SVM(Linear)	0.815	0.813	0.820	0.739	0.860	0.869	0.828	0.862	0.779	0.840	0.819	0.820
	SVM(RBF)	0.731	0.727	0.918	0.898	0.899	0.894	0.867	0.879	0.815	0.825	0.799	0.798
	MLP	0.769	0.768	0.833	0.758	0.887	0.882	0.859	0.874	0.815	0.832	0.802	0.803
	Ours	**0.823**	**0.825**	**0.922**	**0.959**	**0.848**	**0.822**	**0.828**	**0.787**	**0.779**	**0.783**	**0.841**	**0.841**
$\mathcal{HGFS}$	**DT**	0.718	0.714	0.751	0.300	0.537	0.469	0.719	0.711	0.636	0.642	0.645	0.645
	RF	0.825	0.823	0.714	0.695	0.840	0.830	0.883	0.900	0.785	0.836	0.806	0.806
	Adaboost	0.766	0.762	0.740	0.731	0.728	0.698	0.828	0.846	0.764	0.815	0.760	0.760
	SVM(Linear)	0.817	0.815	0.848	0.779	0.911	0.907	0.898	0.899	0.821	0.840	0.838	0.838
	SVM(RBF)	0.807	0.804	0.528	0.365	0.813	0.793	0.898	0.903	0.826	0.846	0.761	0.762
	MLP	0.793	0.790	0.868	0.807	0.918	0.916	0.945	0.946	0.826	0.841	0.831	0.831
	Ours	**0.819**	**0.821**	**0.924**	**0.961**	**0.946**	**0.948**	**0.898**	**0.896**	**0.846**	**0.821**	**0.859**	**0.859**
$\mathcal{HGFSB}$	**DT**	0.718	0.714	0.751	0.300	0.537	0.469	0.719	0.711	0.636	0.642	0.645	0.645
	RF	0.837	0.835	0.738	0.627	0.809	0.798	0.891	0.908	0.780	0.840	0.813	0.815
	Adaboost	0.766	0.762	0.740	0.63	0.728	0.698	0.828	0.846	0.764	0.815	0.760	0.760
	SVM(Linear)	0.819	0.817	0.874	0.816	0.907	0.902	0.883	0.890	0.779	0.801	0.839	0.839
	SVM(RBF)	0.758	0.754	0.935	0.904	0.907	0.903	0.875	0.895	0.795	0.849	0.817	0.816
	MLP	0.807	0.804	0.716	0.875	0.914	0.912	0.898	0.905	0.805	0.814	0.844	0.843
	Ours	**0.849**	**0.851**	**0.952**	**0.976**	**0.934**	**0.938**	**0.891**	**0.885**	**0.862**	**0.845**	**0.881**	**0.881**



(a) (b) (c) (d) (e)

Fig. 6. Feature Importance of *Perspicio* for Different Scenes.

D. Comparison With Related Works

We conduct a comprehensive evaluation of related works using a high-level privacy analysis framework for online social network (OSN) photo sharing proposed in [49]. As shown in Table II, compared to several carefully selected related approaches, our *Perspicio* achieves more privacy-preserving photo sharing goals, providing better privacy protection for users sharing photos on OSNs.

E. Comparisons With Videre

1) Prediction Performance Comparison: We conduct a comprehensive comparison and evaluation of *Videre* and our *Perspicio*. In our experiment, for fair comparisons, the results for *Videre* are executed using their proprietary code alongside the dataset we gathered.

Scene Prediction Performance Comparison: Table III provides an in-depth performance comparison of four classifier models across five distinct scenes, evaluating them on Accuracy, Precision, Recall, and F1-Score. The models compared include *Videre* [27], which relies on gaze direction and head orientation; TM_1, an enhanced version of *Videre* with a dynamic weighting mechanism; TM_2, using the same five features as our model but without dynamic elements; and our *Perspicio*, the advanced model that combines the dynamic weighting mechanism with the full feature set. These metrics, resulting from extensive training and validation, reflect each model's effectiveness in scene classification.

TABLE II
COMPARISON OF REPRESENTATIVE RELATED WORKS

Category	Scheme	Target Privacy	Key Technique	E1	E2	E3	E4	E5	E6	E7
User-driven privacy protection	[11]	P1	*Wireless Communication*	○	◐	○	◐	●	●	○
	[12]	P1	*Wireless Communication*	●	●	○	○	●	●	○
	[13]	P1	*Wireless Communication*	●	●	●	○	●	○	○
	[14]	P1	*Wireless Communication*	●	●	○	○	●	●	○
	[17]	P1	*Wireless Communication*	●	●	○	○	●	*N/A*	○
	[16]	P1	*Deep learning*	○	○	○	●	○	*N/A*	○
Server-based access control	[20]	P1	*Rule-based*	●	●	●	○	●	○	○
	[21]	P1	*Human-computer Interaction*	●	●	●	◐	●	●	○
	[22]	P1; P3	*Rule mining*	●	◐	●	◐	●	●	○
	[23]	P1; P3	*Rule mining*	●	◐	●	◐	●	●	○
	[25]	P1; P2; P3	*Rule mining*	○	●	●	◐	○	●	○
	[26]	P3	*Wireless Communication*	○	◐	○	◐	○	*N/A*	○
Machine learning -based techniques	[5]	P1	*Machine learning*	*N/A*	*N/A*	●	*N/A*	*N/A*	*N/A*	○
	[27]	P1; P3	*Machine learning*	○	◐	●	●	○	○	○
	Ours	P1; P3	*Machine learning*	●	●	●	●	●	●	●

● = True; ◐ = Partially True; ○ = False. P1: observable privacy; P2: inferential privacy; P3: contextual privacy. E1–E6 denote the six functionality goals of privacy-preserving photo sharing in [49], namely privacy-utility tradeoff, personalization, independence, automation, flexibility, and communication-effectiveness. E7 denotes the functionality of supporting individuals' privacy protection outside the systems.

TABLE III
COMPARISON OF ACCURACY, PRECISION, RECALL, AND F1-SCORE METRICS OF FIVE SCENES FOR TRAINED CLASSIFIER MODELS

Approach	Accuracy	Precision	Recall	F1-Score
Daily Life				
Videre	0.805	0.805	0.805	0.803
$\boldsymbol{TM_1}$	0.820	0.840	0.820	0.822
$\boldsymbol{TM_2}$	0.807	0.807	0.807	0.804
Perspicio	**0.849**	**0.861**	**0.849**	**0.851**
Art				
Videre	0.494	0.750	0.494	0.326
$\boldsymbol{TM_1}$	0.465	0.465	0.465	0.635
$\boldsymbol{TM_2}$	0.916	0.923	0.916	0.875
Perspicio	**0.952**	**0.952**	**0.952**	**0.976**
Meeting				
Videre	0.817	0.809	0.817	0.798
$\boldsymbol{TM_1}$	0.751	0.881	0.751	0.786
$\boldsymbol{TM_2}$	0.914	0.911	0.914	0.912
Perspicio	**0.933**	**0.949**	**0.934**	**0.938**
Hosptial				
Videre	0.875	0.880	0.875	0.877
$\boldsymbol{TM_1}$	0.914	0.915	0.914	0.915
$\boldsymbol{TM_2}$	0.898	0.919	0.898	0.905
Perspicio	**0.890**	**0.885**	**0.891**	**0.885**
Public Transit				
Videre	0.867	0.874	0.867	0.869
$\boldsymbol{TM_1}$	0.856	0.852	0.856	0.850
$\boldsymbol{TM_2}$	0.805	0.830	0.805	0.814
Perspicio	**0.862**	**0.876**	**0.862**	**0.845**

TM_1: *Videre* with Dynamic Weighting Mechanism;
TM_2: *Videre* with Five Features Used in Our Model.

TABLE IV
RUNTIME OF FEATURE EXTRACTION

Features	Emotion	Scene	Posture	Gaze	Head
Times	6.96 ms	109.2 ms	30.45 ms	11.47 ms	61.56 ms

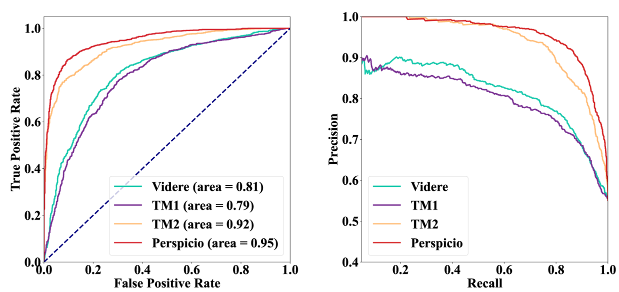
Our analysis offers critical insights into each model's performance. In the Daily Life scene, our *Perspicio* demonstrates high precision and recall, indicating its adeptness at interpreting daily activities. This model also excels at the Artistic Photo scene, significantly outperforming the basic *Videre* model, which struggles with the complexity of artistic contexts. The value of the dynamic weighting mechanism is particularly evident in the Meeting and Hospital scenes, where our *Perspicio* and TM_1 surpass *Videre*. Notably, TM_1 achieves outstanding Accuracy and Precision in the Hospital scene, highlighting the effectiveness of the dynamic weighting mechanism in structured settings. In the Public Transit scene, our *Perspicio* maintains strong performance, but the competitive results from *Videre* suggest that basic features can be effective for certain scene classifications. Interestingly, TM_2 occasionally outperforms TM_1, especially in the Artistic Photo scene, indicating that a comprehensive feature set can sometimes be more advantageous than dynamic elements. From these findings, it is evident that our *Perspicio*, with its integrated dynamic weighting mechanism and extensive feature set, consistently delivers superior performance across various scenes. This suggests that its design is not only versatile but highly capable of navigating the complexities inherent in different contexts. The relative underperformance of *Videre* in most scenes suggests that while basic features are useful, they alone are not sufficient for intricate scene classification. Our study underscores the importance of combining dynamic weighting mechanism with a comprehensive feature set, as demonstrated by our *Perspicio*'s exemplary performance, establishing it as a leading model for effective scene classification.

Comprehensive Prediction Performance Comparison: Table VI offers a thorough performance comparison between the

TABLE V
RUNTIME(MS) OF EACH PHASE OF MULTI-FEATURE ATTENTION INTEGRATOR

Schemes	FN	FE	DW	Classification	Total
SVM(Linear)	2	73.02	-	3908.55	3981.57
SVM(RBF)	2	73.02	-	7116.67	7189.69
Videre	2	73.02	-	16.93	89.95
SVM(Linear)	5	219.56	-	7772.55	7992.11
SVM(RBF)	5	219.56	-	19538.62	19758.18
Perspicio	5	219.56	244.61	32.03	496.19

FN: Feature Number; FE: Feature Extraction; DW: Dynamic Weighting.



(a) ROC Positive Rate (b) Recall

Fig. 7. Receiver Operating Characteristic (ROC) and Precision vs. Recall (PR) Curves of *Videre*, *Perspicio*, and the Transition Models.

Videre and our *Perspicio* across two key conditions: awareness and unawareness. This comparison evaluates Accuracy, Precision, Recall, and F1-Score, considering different feature sets and the inclusion of dynamic weighting (DW) in these models. Overall, our *Perspicio*, equipped with dynamic weighting and a comprehensive feature set ($\mathcal{HGFSB}$), outperforms *Videre* across all performance metrics. This highlights our *Perspicio*'s efficacy in utilizing a full range of features to achieve high predictive accuracy. When analyzing condition-specific performance, our *Perspicio* consistently surpasses *Videre* in precision, recall, and F1-score under the awareness condition, demonstrating its adeptness in scenarios where subjects are cognizant of being observed. The model's superiority is even more evident in the unawareness context, where our *Perspicio* exhibits exceptional precision and recall, culminating in the highest F1-score among all models and conditions. This underscores its remarkable performance in scenarios where subjects are oblivious to observation, a notable challenge for many models. The progression observed in TM_1 and TM_2 when dynamic weighting is applied is noteworthy. Specifically, TM_2 benefits greatly from incorporating facial expression, scene, and body language features alongside dynamic weighting, significantly enhancing its performance over the basic *Videre*, which lacks dynamic weighting and features a narrower set of features.[2]

Prediction Time Comparison: In the time analysis presented in Table IV, emotional feature extraction emerges as the most efficient, requiring a mere 6.96 milliseconds, in stark contrast with the scene feature extraction, which is the most time-intensive at 109.2 milliseconds. As delineated in Table V, our *Perspicio* incorporates dynamic weighting, culminating in an aggregate runtime of 496.19 ms. We can achieve noticeable optimization by extracting multiple features in parallel within the system. This shows that it is quite efficient when compared with SVM-based models. *Videre* is also fast because it uses fewer features. Our approach optimizes between sophisticated feature integration and runtime, maintaining practicality for real-time application scenarios.

Receiver Operating Characteristic and Precision vs. Recall: Receiver Operating Characteristic (ROC) curves are a foundational method for evaluating the performance of binary classification models through probabilistic predictions. The model's accuracy is quantified by the Area Under the Curve (AUC), where a value close to 1 indicates high accuracy, while a value close to 0.5 suggests that the model's predictions are no better than random chance.

Fig. 7 depicts the ROC and Precision vs. Recall (PR) curves for *Videre*, our *Perspicio*, and Transition Models (TM_1 and TM_2). These curves offer a quantitative metric for evaluating the classification efficacy of each model. The area under ROC curve (AUC) for our *Perspicio* is noted at 0.95, indicating a high degree of classification performance. *Videre* follows with an AUC of 0.81, and TM_1 is observed with an AUC of 0.79. These values may reflect the generalization potential of our *Perspicio* within the tested dataset. The PR curve analysis further reveals that our *Perspicio* achieves an optimal balance between precision and recall, positioning it near the upper-right quadrant of the PR space. This suggests that our *Perspicio* is capable of maintaining high precision while achieving significant recall levels, a characteristic that is often sought after in classifier models.

The comparison indicates that while TM_1 benefits from a dynamic weighting mechanism, its absence of a complete feature set appears to limit its performance. Conversely, TM_2 shows enhanced performance over *Videre* due to its inclusive feature set, yet it does not exhibit the same level of adaptability as our *Perspicio*. These findings point to the conclusion that comprehensive feature integration, along with an adaptable modeling approach, is crucial for the development of advanced classification models. Such models are necessary to handle complex and varied datasets effectively. This insight highlights the importance of both an extensive feature set and adaptive model capabilities, which should be a focus in the continued development of sophisticated classification models for complex tasks.

2) Authorization Mechanism Performance Comparison: In this section, we evaluate the performance of the proposed authorization mechanism and compare with *Videre*.

Computation Overhead: In our *Perspicio*'s photo sharing process, the computational overhead is notably governed by the encryption step in the offline phase, as well as the verification during the online phase. We carefully examine the computation cost associated with the main algorithms in these phases, as shown in Table VII. In the scenario where N individuals are authorized for photo sharing, our findings indicate that the time it takes to create a signature for a single user in our system is remarkably quick at 0.05 ms. This is significantly more efficient compared with *Videre*, which requires a total of 9.3 ms. Moreover, our verification time per user for authorization during the online phase is only 0.09 ms, in sharp contrast with

[2]More evaluation about confusion matrices will be given in the Appendix, available online.

TABLE VI
COMPARISON OF ACCURACY, PRECISION, RECALL, AND F1-SCORE METRICS BETWEEN *VIDERE* AND *PERSPICIO*

Schemes	Features	DW	Overall				Awareness			Unawareness		
			Accuracy	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Videre	$\mathcal{HG}$	✗	0.750	0.750	0.748	0.745	0.760	0.643	0.697	0.741	0.835	0.785
TM_1	$\mathcal{HG}$	✓	0.758	0.758	0.758	0.757	0.752	0.690	0.721	0.763	0.814	0.788
TM_2	$\mathcal{HGFSB}$	✗	0.839	0.839	0.839	0.839	0.821	0.821	0.821	0.854	0.853	0.854
Perspicio	$\mathcal{HGFSB}$	✓	**0.882**	**0.882**	**0.881**	**0.881**	**0.851**	**0.891**	**0.871**	**0.908**	**0.873**	**0.890**

DW: Dynamic Weighting; TM_1, TM_2: The symbols have the same meaning as in Table III;
$\mathcal{H}$: Head Orientation; $\mathcal{G}$: Gaze Direction; $\mathcal{F}$: Facial Expression; $\mathcal{S}$: Scene; $\mathcal{B}$: Body Language.

TABLE VII
RUNTIME OF END-TO-END AWARENESS DETECTION

Schemes	Off-line (ms)				
	IDGen	Sign	Agg	Dec	Total
Videre	960	9.3	$0.003n$	-	969.3
Perspicio	-	**0.05**	-	**0.63**	$\mathbf{0.68(1425\times)}$
Schemes	**On-line (ms)**				
	Ver	IDRec	Enc		Total
Videre	$2.69n$	$980n$	-		$982.69n$
Perspicio	$\mathbf{0.09n}$	-	$\mathbf{1.05n}$		$\mathbf{1.14n(862\times)}$

TABLE VIII
COMMUNICATION OVERHEAD COMPARISON

Reuse (byte)	pk_{sig}	sk_{sig}	pk	sk	Total
Videre	128	64	-	-	192
Perspicio	32	32	32	32	128
Each Time (byte)	σ	ct	Img		**Total**
Videre	128+20/n	-	I		128+20/n+I
Perspicio	64	CFI	I		64+CFI+I

IMG: Image Size; CFI: Ciphertext of Facial Image.

Videre's 2.69 ms, demonstrating that an impressive improvement in efficiency. Most of the additional processing time in our algorithm arises from encrypting an image of 16.81 KB in the online phase, which takes 1.05 ms. When combined with the offline decryption time of 0.63 ms, it leads to a total computation time that is practical for real-world applications.

In the offline phase, the total runtime for a single user in our scheme is 0.05 ms, which is 1,425× faster than *Videre*. In the online phase, the total runtime for a single user in our scheme is only 1.05ms, which is 862× faster. The computational efficiency improvements in both phases, by about three orders of magnitude, are due to our adoption of a more commonly used trusted third-party setting. This not only significantly enhances privacy protection efficiency but also improves security. The data presented in Table VII strongly confirms that the overhead of our *Perspicio* is both reasonable and suitable for practical deployment.

Communication Overhead: Our *Perspicio*'s communication overhead incorporates public and private keys, signatures, and ciphertexts, which are essential for secure photo sharing. Referencing Table VIII, we detail the overhead associated with our *Perspicio*, setting security parameter to 256 bits. The system adopts ECDSA for digital signatures, generating shorter public and private keys through ECC, especially for reusable keys of size 32 bytes each. These keys can be efficiently reused across multiple photo sharing sessions. For each instance of photo sharing, the data to be exchanged includes the signature σ, the encrypted facial data ct, and the photo Img. The dominant factor in communication overhead is actually determined by the size of the transmitted image I. Our scheme partially encrypts a photo, which can be roughly viewed as the original photo plus the ciphertext of the facial region. Since the encryption algorithm is AES, the total communication overhead of our scheme is almost the same as that of *Videre*.

VII. RELATED WORK

Research on privacy-preserving photo sharing can be broadly divided into three main categories: user-driven privacy protection, server-based access control, and computer vision-based automated detection.

The first category, *user-driven privacy protection*, involves mechanisms where the subject of a photo collaborates with the photographer to ensure privacy when sharing images. Besmer and Lipford [10] examined privacy concerns in photo sharing on Facebook, finding that users struggle with ownership and display control due to tagging features, complicating identity management. To address this, they introduced a tool that allows users to restrict photo visibility at the time of posting, though it places a significant manual burden on users. Similarly, Xu et al. [15], [18], [19] proposed a system where all individuals in a photo are involved in sharing decisions, leveraging facial recognition (FR) technology. They developed a distributed collaborative learning method to create personalized FR systems, reducing computational complexity while maintaining privacy. Zhang et al. [12] introduced the COIN system, enabling users to set privacy preferences and allowing service providers to enforce them, even erasing unwanted individuals from photos. Other notable contributions include Aditya et al.'s I-Pic platform [11], which respects user-chosen privacy levels, and Ra et al.'s Do Not Capture (DNC) technology [14], which blurs the faces of individuals who opt out of being photographed. Li et al. [13] developed PrivacyCamera, which automatically notifies and protects strangers in the background of photos. Although effective, these solutions rely on cooperation from both parties and are vulnerable to malicious actors who choose not to use designated apps.

The second category, *server-based access control*, focuses on centralized methods to mitigate privacy risks. Early research highlighted privacy threats due to inadequate access control over social networks [50], leading to more flexible access control models based on social contexts [51], [52]. However, these models still lack the requirement for explicit permission from individuals captured in photos. Squicciarini et al. [28] introduced a collaborative game-theoretic approach where both photo owners and co-owners must agree on privacy settings, though co-owners are often identified through tagging systems. Henne et al. [20] proposed SnapMe, which notifies users when photos taken in predefined zones are uploaded. QR codes and tagging systems, as introduced by Bo et al. [21] and Tonge et al. [26], allow users to opt out of photo sharing, though their use is limited by the need for physical markers. Other approaches, such as Ilia et al.'s [22] fine-grained control over personally identifiable information (PII) and Vishwamitra et al.'s [23] collaborative privacy management, offer more detailed controls but are often cumbersome. Li et al. [25] introduced HideMe, which automates privacy enforcement through context-based policies, though these solutions are limited by the unpredictability of photo capture.

The final category, *computer vision-based automated detection*, provides a more scalable approach. Hasan [5] and Zhang [27] both developed machine learning classifiers capable of automatically detecting bystanders and unaware individuals in photos, respectively. These methods represent a significant step forward in addressing non-consensual photo sharing by leveraging advanced image recognition technologies.

In comparison with Zhang's work [27], our research aims to improve detection accuracy and applicability across different scenarios. We further propose a lightweight authorization mechanism that efficiently protects users' privacy during consent-seeking for photo sharing, a feature not present in *Videre*. The workflow of our *Perspicio* integrates with existing frameworks, such as [5], [27], and others in this field.

A. Limitations

Non-consensual photo sharing still has several unresolved issues that deserve further investigation. The first major challenge is the difficulty of guaranteeing 100% detection accuracy across all scenarios. While our *Perspicio* system exhibits high detection accuracy in diverse environments, it is not immune to detection errors. Previously unseen or insufficiently represented scenarios may still lead to lower accuracy, which could in turn risk inadvertent privacy leakage, especially for users not registered with the system.

Second, *Perspicio* may not fully protect against all adaptive attacks. For example, a photographer may maliciously edit a photo so that the system fails to detect a person, thereby preventing consent inquiry. Such manipulation could result in unintended photo-privacy leakage. Similarly, malicious photo editing may also reduce the accuracy of awareness classification, further increasing the privacy risk for unregistered individuals. As a possible mitigation, JPEGsnoop-like technology [53] can be incorporated into the photo pre-processing stage to detect tampered photos and help preserve integrity.

Another limitation is that whether a person is "aware" of being photographed is inherently subjective and context-dependent. In this work, the corresponding labels are constructed as an operational approximation based on observable visual and contextual cues, following a common practice adopted in prior studies such as Videre and related works. While this formulation is useful for building and evaluating a deployable system, it may still be affected by annotator subjectivity, conflicting visual evidence, ambiguous cases, and uncertainty in borderline samples. Although our multi-feature design and dynamic weighting mechanism help mitigate part of this ambiguity, they do not fully resolve annotation reliability, adjudicated ambiguity handling, or uncertainty-aware benchmark construction. In particular, the current version should not be interpreted as providing the strongest possible validation in terms of formal agreement statistics or explicitly curated uncertainty-aware subsets. A more rigorous treatment of these issues would further strengthen the task formulation and improve the robustness of the overall system, which forms as important directions for future work.

Other limitations, such as robustness in complex environments, liveness detection, two-dimensional image constraints, ambiguity in social contexts, storage centralization, and broader ethical considerations, are discussed in Appendix IX-G, available online for completeness.

VIII. Conclusion

In this paper, we presented *Perspicio*, a privacy-preserving non-consensual photo sharing system, poised to address the accuracy, efficiency, and privacy challenges. Our *Perspicio* employs an innovative feature fusion strategy and dynamic weighting mechanism, significantly improving the detection accuracy of individuals in social media images. Throughout its development and evaluation, our *Perspicio* has shown substantial improvements in prediction accuracy over *Videre*. It has introduced an efficient authorization process, giving control over facial data back to the users and bolstering their privacy. This approach has not only sped up processing but also addressed users' photo privacy concerns. Since our system can automatically detect unaware individuals in complex environments solely depending on image data, we believe that *Perspicio* demonstrates promising prototype-level feasibility for privacy-aware photo sharing over social networks.

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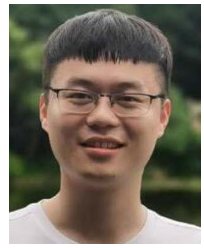

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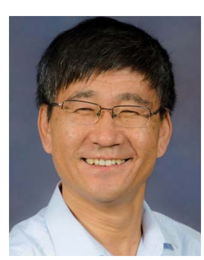

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